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A Tutorial on Cepstrum and LPCCs

The Cepstrum is a sequence of numbers that characterise a frame of speech. The cepstrum computed from the periodogram estimate of the power spectrum can be used in pitch tracking, while the cepstrum computed from the AR power spectral estimate were once used in speech recognition (they have been mostly replaced by [MFCCs](#)).

One of the benefits of cepstrum and LPCCs over e.g. LPCs is that you can do cepstral mean subtraction (CMS) on cepstral coefficients to remove channel effects.

Feel free to leave any comments or queries in the comment section down the bottom of this page.

What is the Cepstrum?

This section will describe the cepstrum computed from the periodogram estimate of the power spectrum. We will first cover autocorrelation a bit, then we will show how the cepstrum is computed in a similar manner. Further on in this article we will cover LPCCs.

The cepstrum can be thought of as being similar to the [autocorrelation sequence](#). If we have the power spectrum of a signal, we can compute the autocorrelation sequence using the [Wiener-Khinchin theorem](#).

In the maths that follows, $x(n)$ is the time domain signal, $X(k)$ is the complex spectrum, $P(k)$ is the power spectrum of $x(n)$ and $A(n)$ is the autocorrelation sequence of $x(n)$.

By taking the Discrete Fourier Transform of $x(n)$ we get the complex spectrum:

$$DFT(x(n)) \rightarrow X(k)$$

If we then take the Inverse Discrete Fourier Transform we get $x(n)$ back again:

$$IDFT(X(k)) \rightarrow x(n)$$

So far nothing amazing.

BUT, by taking the square of the absolute of the DFT of $x(n)$, we get the power spectrum.

$$|DFT(x(n))|^2 \rightarrow P(k)$$

If we take the IDFT of the power spectrum we actually get the Autocorrelation sequence, instead of the original sequence (This is what the [Wiener-Khinchin theorem](#) states):

$$IDFT(P(k)) \rightarrow A(n)$$

Where does the Cepstrum fit into all this? If we take the log of the power spectrum before the IDFT we actually get the cepstrum:

$$IDFT(\log(P(k))) \rightarrow C(n)$$

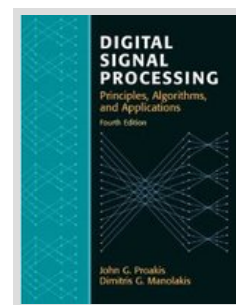
So you can think of the Cepstrum as a log compressed autocorrelation sequence, as it carries information similar to the autocorrelation sequence (about periodicities in $x(n)$), but it is computed from the log power spectrum instead of the standard power spectrum.

Contents

- [What is the Cepstrum?](#)
- [What about LPCCs?](#)
- [Computing LPCCs from LPCs](#)

Further reading

We recommend these books if you're interested in finding out more.

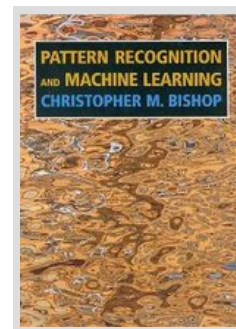


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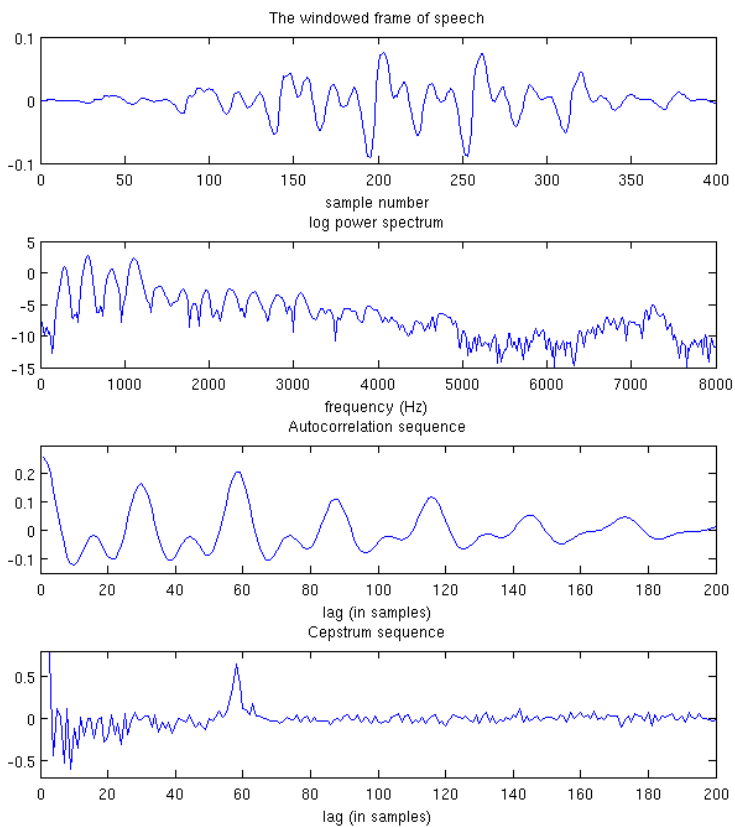


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Plot showing a frame of speech along with the log power spectrum, autocorrelation sequence and cepstrum.

In the figure above you can see the original frame of speech, along with its log power spectrum. The autocorrelation sequence and the cepstrum are also shown. The peak in the cepstrum corresponds to one of the peaks in the autocorrelation sequence, but the peak is much clearer in the cepstrum. This peak is at a lag of 58 samples, this corresponds to a pitch frequency of $(16000/58) = 275\text{Hz}$ (for a 16kHz signal). This is quite a high pitch frequency, the original speech is from a female speaker. The strong peak is the reason why the cepstrum is often used in pitch tracking applications.

To get the cepstrum in matlab, use the following code:

```
PowerSpectrum = abs(fft(SpeechFrame,1024)).^2;
AutoCorrelation = ifft(PowerSpectrum,1024);
Cepstrum = ifft(log(PowerSpectrum),1024);
```

What about LPCs?

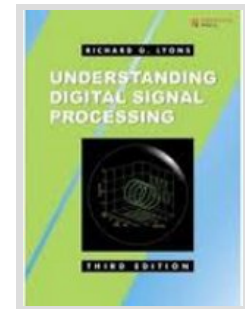
In the previous section we looked at the standard cepstrum, Linear Prediction Cepstral Coefficients are computed in the same way except they are computed from the smoothed Auto-Regressive power spectrum instead of the periodogram estimate of the power spectrum. For an order 10 AR spectral estimate, the Levinson Durbin algorithm is applied to the first 10 Autocorrelation coefficients to compute 10 linear prediction coefficients. In MATLAB this is done using:

```
[lp,g] = lpc(frame,10)
```

The linear prediction coefficients for the frame in the previous section are:

```
1.00 -2.22 1.68 0.05 -1.28 1.32 -0.30 -0.76 1.35 -1.19 0.44
```

To compute the cepstrum from the AR spectral estimate, use the following steps:



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```
[lp,g] = lpc(SpeechFrame,10);
ARPowerSpectrum = g ./ abs(fft(lp,1024)).^2;
Cepstrum = ifft(log(ARPowerSpectrum),1024);
```

Note that the procedure above is exactly the same as the previous section, only the method for computing the power spectrum is different. The first 10 cepstrum are:

```
-11.78  2.22  0.79 -0.12  0.38  0.03 -0.20  0.04 -0.42 -0.11
```

Computed in this way the first coefficient is ignored, it depends only on g . These are the Linear Prediction Cepstral Coefficients for the frame of speech. If you actually want LPCCs though, don't do it this way, as there is a more efficient way of computing them described in the next section.

The point of this section is to show the relationship between the cepstrum computed from the periodogram power spectrum and LPCCs which are the same except they are computed from the AR estimate of the power spectrum instead.

Computing LPCCs from LPCs

There is a simple recursive formula for computing Linear Prediction Cepstral Coefficients directly from LPCs without doing any DFTs. In the following equation a_k are the linear prediction coefficients, called lp in the previous section.

$$c(n) = \begin{cases} 0 & n < 0 \\ \ln(G) & n = 0 \\ a_n + \sum_{k=1}^{n-1} \left(\frac{k}{n}\right) c(k) a_{n-k} & 0 < n \leq p \\ \sum_{k=n-p}^{n-1} \left(\frac{k}{n}\right) c(k) a_{n-k} & n > p \end{cases} \quad (1)$$

From a finite number of LPC coefficients, an infinite number of cepstral coefficients can be calculated. Research has shown, however, that 12–20 cepstral coefficients are sufficient for speech recognition.

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Z

zohaib anwar

9 years ago

I have some question regarding difference between LPCC and MFCC. I want to know which which technique is better in terms of feature extraction and why its better. Secondly can you please explain about the autocorrelation and LPC analysis based on Levinson Durbin algorithms steps in LPCC. If i am not wrong first three steps are same as in MFCC ie pre-emphasis, framing and windowing. I did not understand the last two steps in LPCC and why we apply these steps. Please guide me regarding this.

Thanks

5 2 Reply 

S

Sourya Dey

a year ago

Hi, what is the 'G' when computing LPCC coefficient for $n=0$?

0 0 Reply **Kiara**

7 years ago

I have a question too: I am studying this for the first time and in class my teacher said that you actually have to compute *both* the LPCs and the MFCCs for speech recognition (later with HMMs). But I can't find any information on how to combine them into one feature vector :(

0 0 Reply **Max Ogaz** Kiara

7 years ago

I'm seeking for the same info..

0 0 Reply 

J

jameslyons Mod Max Ogaz

7 years ago

You would not normally use both lpcs and mfccs in the same feature vector. But if you want to, you just stick them together e.g. 12 mfccs and 12 lpcs per frame of speech would result in a combined set of 24 features per frame. Nothing fancy is required to combine them, the classifier will learn their relationship.

0 0 Reply 

D

Dewi

8 years ago

LPCC and LFCC is same or isnt?

0 0 Reply 

comments powered by Disqus



8 years ago

ybl krq 101 k111h r
- tfqzstdsh

Its going to depend on who wrote it, but i would guess lfcc is a linear frequency cepstral coefficient which would be identical to a mfcc except with a linear frequency scale instead of a mel warped frequency scale. Lpccs are different.

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Franz Anders

9 years ago

First of all, thanks for this excellent explanation. I would like to know your source on how to calculate the cepstrum, because in some sources the absolute spectrum is used (for example in "Text-to-Speech-Synthesis" by Taylor), and in other sources the power spectrum is used.

0 0 Reply

Questions/Feedback

Notice a problem? We'd like to fix it!

Leave a comment on the page and we'll take a look.



Juda Masela

9 years ago

Thank you very much!!! This really helped

0 0 Reply



Black Hat

10 years ago

Is there a meaningful reason for calculating the cepstrum from $\log(P(k))$ instead of $P(k)$?

0 0 Reply

S

shasasa → Black Hat

9 years ago

Yes! there is. For that you need to delve into homomorphic processing. In that, we have to convert a non linear system into a linear one, which is what the log operation does, right? Log converts multiplication, into addition, and so in that way ,is imperative.

0 0 Reply



Black Hat

10 years ago edited

Quote: "The peak in the cepstrum corresponds to one of the peaks in the autocorrelation sequence, but the peak is much clearer in the cepstrum"

Does the peak in the cepstrum sequence always correspond with the second peak in the autocorrelation sequence?

0 0 Reply

J

jameslyons Mod

→ Black Hat

10 years ago

Nope, it will often correspond to the largest peak, but neither of these methods are foolproof, and they'll occasionally give you wrong answers. It is just that cepstrum usually works a bit better on speech signals.

0 0 Reply



Black Hat

→ jameslyons

10 years ago

Thanks - what was a quick response btw. So if the fundamental frequency is suppressed (happens a lot in birdsongs - which I'm working on), then the peak will correspond to one of the harmonics, correct?

0 0 Reply

J

[jameslyons](#)

Mod

[Black Hat](#)

10 years ago

That is most likely yes

0

0

Reply

[Black Hat](#)[jameslyons](#)

10 years ago

Thanks. Great writing btw - very easy to understand.

0

0

Reply



S

[Srikanth Kallakuri](#)

11 years ago

That was really awesome. Thank you very much.

0

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